

Evaluating the Claim that Preregistration and Registered Reports Restrict Exploratory Research

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Re-Submission for *Lifecycle Journal*

Version Date: April 30, 2026

Declarations

Funding: No specific funding was received in support of this research. A small stipend was provided by the Center for Open Science for submitting this research plan prior to a specified deadline.

Open data: All data, code, and materials will be made openly available at <https://osf.io/4vbtq/>

Conflicts of interest: None to declare

Ethics: This is an archival study of published research and does not involve human subjects.

GenAI: No Generative AI or LLMs were used by the authors in preparation of this manuscript. PaperWizard and ReviewerZero were two AI-based evaluation services used by Lifecycle Journal, and the manuscript was revised based on the feedback provided.

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Abstract

A persistent concern about implementation of preregistration and Registered Reports in psychology is that doing so would reduce the frequency and value of exploratory research. As we are nearly 15 years on from the initial proposal to adopt registration in psychology, it seems time to formally examine whether these concerns have any merit. The purpose of the present study is to examine the frequency of reports of exploratory research in registered and non-registered articles. In this proposed project, we replicate and extend a study by O'Mahony (2023) by examining a matched set of Registered Reports, preregistered articles, and traditional articles (total $N = 300$) for the frequency of reported exploratory research, defined as any analysis that is indicated to be exploratory in nature or that is indicated to not be a planned test of a specific hypothesis. This project will provide strong data that should provide an empirical basis to be used in future discussion about the impact of registration on the reporting of exploratory research.

Keywords: *exploratory research, preregistration, registered reports, metascience, open science*

Psychology's replication crisis has led to a host of new policies and practices meant to improve the state of our science (Munafò et al., 2017). A practice that is central to the reform movement is *preregistration*, which involves creating a supposedly accessible and unalterable record of a study plan prior to collecting data and/or conducting analyses (Nosek et al., 2018). Rooted in the clinical trial registry previously developed for medical research (Boutron et al., 2016), preregistration is a practice that supports transparency about researchers' plans, and facilitates comparisons between how those plans correspond to what they actually report (Claesen et al., 2021; Willroth & Atherton, 2024). An early and persistent concern about widescale implementation of preregistration in psychology is that doing so would reduce the frequency and value of exploratory work, and therefore restrict creativity, serendipity, and discovery (Goldin-Meadow, 2016; Scott, 2013). As we are nearly 15 years on from the initial proposal to adopt preregistration in psychology, it seems time to formally examine whether the concern about preregistration restricting exploratory work has any merit. Accordingly, the purpose of the present study is to examine the frequency of reports of exploratory research in registered and non-registered articles.

Fallout from the Confirmatory/Exploratory Divide

The function that preregistration is meant to serve is not always clear or consistent (Lakens, 2019). Resolving debates about the function of preregistration is outside of the scope of the current project, as are debates about the utility of preregistration (see Lakens, 2019; Syed, 2024; Szollosi et al., 2020 for discussion of these issues). An early prominent argument for preregistration is that it allows for a clearer separation between *confirmatory* analyses, or *hypotheses generated prior to the study*, and *exploratory* analyses, or *analyses for which there were no prior hypotheses* (Wagenmakers et al., 2012). This framing was in direct response to concerns about questionable research practices (or *p*-hacking, researcher degrees of freedom, garden of forking paths; Gelman & Loken, 2013; Head et al., 2015; Simmons et al., 2011), in which researchers fail to disclose all of the analyses that were conducted other than the statistically significant results presented in their papers, as well as HARKing (hypothesizing after results are known; Kerr, 1998), in which researchers reframe exploratory findings as though they were confirmatory.

The distinction between confirmatory and exploratory is not always so clear in practice (Szollosi & Donkin, 2021; Wagenmakers et al., 2012). Part of the problem is that there is not a clear or consistent definition of what constitutes exploratory research. Indeed, this is an active area of discussion (Feest & Devezer, 2025). Lack of clarity is partially related to the fact that confirmatory/exploratory forms a continuum of potential data analytic approaches (Wagenmakers et al., 2012). Accordingly, many have moved away from emphasizing the distinction in the context of preregistration in favor of a distinction between data-independent and data-dependent decision-making (e.g., Srivastava, 2018). Nevertheless, the confirmatory/exploratory distinction was prominent as preregistration was gaining momentum. Despite proponents of preregistration arguing to the contrary (Chambers, 2013; Wagenmakers et al., 2012), there were quickly concerns expressed that preregistration would both *restrict* and *devalue* exploratory research, with exploratory in this context meaning *any analysis that is indicated to not be a planned test of a specific hypothesis*. The message about the utility of preregistration was seemingly taken by many to mean that only confirmatory research via preregistration was research worth doing.

Registered Reports are a more restrictive version of preregistration that embeds the practice into the process of peer review (Chambers, 2013), and similar concerns about the format restricting exploration have been made (see Besançon et al., 2021). The review process for Registered Reports

is separated into two distinct sections: Stage 1, consisting of the Introduction, Method, and Planned Analyses, which are prepared prior to executing the study; and Stage 2, consisting of the previous sections plus the Results and Discussion, submitted for review following completion of the study. Review platforms¹ make acceptance decisions based on the Stage 1 manuscript and review process, addressing the question: will this study produce worthwhile knowledge, regardless of the results? If the answer is yes, then the review platform agrees to accept the full Stage 2 once complete, regardless of the results. Registered Reports are intended to address the problem of publication bias, or the fact that the published literature does not adequately represent the universe of studies completed, but rather studies have been selected via a biased process (most typically in favor of statistically significant findings). Addressing the problem of publication bias reduces the motivation to engage in questionable research practices, HARKing, and other unscientific behaviors (Chambers, 2013).

The formal guidelines for Registered Reports specify the need to make a very clear distinction between confirmatory and exploratory analyses, because anything not specified as hypothesized in the Stage 1 manuscript is considered to be exploratory. Importantly, there is absolutely no restriction on including exploratory analyses within Registered Reports. Researchers can include planned exploratory analyses in their Stage 1 proposal and are free to add whatever they wish at Stage 2, within reason, so long as they are clearly labeled as exploratory and are not given undue weight when interpreting the full findings of the study (Chambers & Tzavella, 2022).

Nevertheless, Registered Reports are confirmatory-focused, and indeed use language and procedures that are much more aligned with quantitative hypothesis-testing approaches compared to other approaches. That said, Registered Reports can and have been used with a wide variety of research designs, including qualitative studies (Karhulahti et al., 2023), secondary data (Davis-Kean et al., 2024), meta-analyses (Kathawalla & Syed, 2021), and many more. Concerns about the heavy focus on confirmatory approaches led to the development of a companion format, Exploratory Reports, wherein researchers largely eschew inferential statistics in favor of extensive descriptive statistics and data visualizations (McIntosh, 2017). Despite the fact that they were introduced many years ago, Exploratory Reports have been taken up by few review platforms and are seldom used.

Taken together, both preregistration and Registered Reports seek to improve the transparency and clarity with which researchers describe the epistemic status of their work. The intention is *not* to favor one form of inquiry over another, although we recognize that in practice this could very well be what happens. Rather, the intention is to support clearer delineation of confirmatory/planned and exploratory/unplanned analyses via a dedicated subsection of the Results section or clear and specific language that denotes an analysis as exploratory. Despite many proponents of preregistration being clear about this intention (Wagenmakers et al., 2012), concerns persist (McDermott, 2022). As an ostensibly empirical science, however, at some point concerns need to be adjudicated via data.

The most directly relevant data of which we are aware comes from a recent Ph.D. thesis from O'Mahony (2023). Embedded within a much larger examination of similarities and differences between Registered Reports ($n = 170$) and traditional articles ($n = 340$), the findings indicated that

¹ We use the term “review platforms” instead of the more common “journals” in recognition of the growing prominence of journal-independent review platforms, such as PCI Psychology (Bottesini et al., 2025) and MetaROR (MetaROR, n.d.). Thus, “review platforms” is a more inclusive term for the current landscape of scientific dissemination.

exploratory research was in fact *more* commonly identified in the Registered Reports (75%) compared with the traditional articles (57%). This finding runs counter to the many critics claiming that preregistration would *reduce* the frequency of exploratory research. Additionally, Soderberg et al. (2021) found that Registered Reports and traditional reports were statistically indistinguishable on ratings of creativity and novelty, again running counter to critics' claims. Our focus in the present study is to replicate and extend O'Mahony's (2023) study with a narrow focus on the frequency of reported exploratory research across article types.

A major challenge to drawing any conclusions about the frequency of exploratory research between registered and non-registered articles is that the ability to do so is entirely dependent on researchers' reporting (henceforth, we use the term "registration" to refer to both preregistered studies and Registered Reports). Given the widespread prevalence of HARKing, in which exploratory findings are reframed and presented as confirmatory, it is simply not possible to derive an accurate estimate of how often exploratory research is occurring when there are no reporting restrictions in place (Wagenmakers et al., 2018). Thus, when making such comparisons, examining the reporting practices in articles is the best available indicator. Although reports are an imperfect proxy for actual practice, they remain highly relevant to the debate about the status of exploration in registered articles, and specifically address the claims that registration would restrict exploratory research. That is, if registration does in fact restrict exploration, there should be low levels of exploration reported in registered research articles.

The Present Study

The purpose of the present study is to conduct an initial investigation into whether preregistration and Registered Reports restrict investigators' reports of exploratory research, or *any analysis that is indicated to not be a planned test of a specific hypothesis*. We do so by comparing the prevalence of exploratory research reported across articles with preregistered studies, Registered Reports, and traditional articles. The purpose of the study is not to get a population estimate of exploratory research in psychology, but rather to examine the specific claims that registration would lead to a decrease in reported exploratory research.

The competing views on the practice of registration and its relation to exploratory research would lead to two divergent data patterns: If registration restricts exploratory research, then traditional articles should include higher prevalence of reported exploratory research than registered studies (assuming no or low prevalence of HARKing²). In contrast, if registration clarifies the status of exploratory research, then registered studies should include higher prevalence of reported exploratory research than traditional articles. The only empirical examination of this question that we are aware of found *higher* prevalence of exploratory research in Registered Reports compared to traditional reports. Accordingly:

Hypothesis 1: There will be a higher prevalence of reported exploratory research in preregistered articles compared to traditional articles.

Hypothesis 2: There will be a higher prevalence of reported exploratory research in Registered Reports compared to traditional articles.

² This is not a reasonable assumption, however

As noted, the current study will not be able to actually assess the frequency of exploratory research, *per se*, but rather the frequency with which exploratory research is reported and transparently described. Differences in reporting expectations could lead to differences between preregistered articles and Registered Reports. Several studies have reported that researchers often do not adhere to their preregistration plans nor do they transparently disclose deviations from their plans (Claesen et al., 2021; van den Akker et al., 2023; Willroth & Atherton, 2024). Thus, an article could be preregistered but essentially indistinguishable from a traditional article in its reporting. Registered Reports, however, are much more regulated given how the review process operates, and should thus have clear and transparent reporting. Accordingly:

Hypothesis 3: There will be a higher prevalence of reported exploratory research in Registered Reports compared with preregistered articles.

The aforementioned studies will be examined within a set of articles in which all of the reported studies were registered or not registered. Some articles, however, consist of a mix of registered and unregistered studies. The existing data (Liu et al., 2025), along with our own preliminary review, suggest that the number of mixed articles is rather low. Thus, we will conduct exploratory within-article analyses of the frequency of exploratory research reported within mixed registered and non-registered articles.

Finally, we additionally take on some exploratory analyses to examine whether reporting practices have changed over time. Although exploratory, our interest in this question is sparked by several possibilities that we can imagine. First, because of the heightened interest in transparent reporting over the last 15 years, we may see an increase in traditional articles labeling analyses as exploratory. Comparing traditional articles before and after implementation of registration in psychology would address this point.

Second, the early days of registration in psychology may have been associated with less than ideal implementation by authors, reviewers, and editors, due to the fact that it was a new and unfamiliar practice. Accordingly, some may have had more rigid views about what was permissible to include in registered studies. Thus, there could be less exploratory research reported in registered studies when the practices were initially adopted but have subsequently increased over time. We will explore this possibility for both preregistered articles and Registered Reports.

Method

Primary Article Selection Process

The corpus of articles for the primary analysis will consist of three types of articles as detailed below ($n = 100$ of each; total $N = 300$). To reduce heterogeneity, only quantitative articles that include statistical analysis of experimental or observational data will be included. Articles based on qualitative, mixed methods, or meta-analytic approach will be excluded. This exclusion is not because that work is less valuable, but rather because it is less frequently registered than quantitative research.

Articles reporting a preregistered study (“preregistered articles”): An empirical article, a) in which all studies are stated to be preregistered and b) for which the registration document can

be located (either via an included link in the manuscript or via manual search). Because of the wide variety in implementation quality, we will not require that the registration document has been formally registered, as long as it can be located (e.g., a registration document is on the OSF project page as a file, but not as a formal registration).

Articles reporting a Registered Report (“Registered Reports”): An empirical article that is a) stated, or clearly implied, to be a Registered Report and b) for which the accepted Stage 1 manuscript can be located (either via an included link in the published Stage 2 manuscript or via manual search). Because of the wide variety in implementation quality, we will not require that the Stage 1 manuscript has been formally registered, as long as it can be located (e.g., a registration document is on the OSF project page as a file, but not as a formal registration).

Articles with no reported registrations (“traditional articles”): An empirical article that includes no statements or link regarding any form of registration. This group of articles will consist of two sub-groups: a group of articles that is matched to the target registered articles and a second group of articles published in 2010, prior to the adoption of registration in psychology. The year 2010 was selected because it immediately predates major awareness of the replication crisis in psychology, and registration had not yet been introduced to psychology.

Most of the meta-scientific research on preregistration has relied on one of two datasets: articles that were 1) part of the Center for Open Science Preregistration Challenge or 2) included in a public database listing articles that received a preregistration badge. Both are limited: the Preregistration Challenge was a unique experience with specific guidelines and a review process by the Center for Open Science. The badge database includes articles published only up to 2020. Neither is specific to psychology. The reason these limited resources have been used is that identifying a proper set of articles is no easy task. We adopt an approach that we believe is well-suited to our research questions. Following calls to make metascientific data transparent and openly available (Elson et al., 2026), all articles included in the analysis as well as our coding will be posted at <https://osf.io/4vbtq/>. This matched set of articles will have high reuse potential among metascientists.

We will first start with journals that are documented to publish Registered Reports. These journals are a good starting point, because publishing Registered Reports is a strong signal that the journal values registration for empirical studies. Accordingly, if the criticism that registration restricts exploratory work is true, it should be most evident in the registered articles published in these journals. As a check on the implications of using this starting point, we will also identify a small set ($n = 25$) of traditional articles from journals that do not publish Registered Reports. Details about the article selection process for this supplemental set are described below.

Liu et al. (2025) recently published a metascientific study of Registered Reports in psychology that serves as a useful starting point. Through their search process, they identified 47 journals that publish Registered Reports and were classified as psychology journals by the 2023 Clarivate Journal Citation Report (see list at <https://osf.io/jwaf9>). This is not an exhaustive list, as some journals that publish the format were not included³. This is not a problem for the current study, as we are not seeking to generate population estimates of how often exploratory work is

³ e.g., *Comprehensive Results in Social Psychology*, likely excluded at least in part because it only publishes Registered Reports, and thus no companion traditional article could be included for comparison.

published in the literature, but rather the relative prevalence of exploratory work in different article types.

Within these journals, they identified 237 Registered Reports published up to April 1, 2023, with the first published Registered Report appearing in *Cortex* on February 13, 2015⁴. Liu et al. (2025) excluded replication studies from their set, as well as articles for which they could not identify a match, and one article that had been retracted, bringing the total for analysis to 119. We also chose to exclude replication studies from the current project, as they have a different function from original research and thus could show disparate patterns. We are, however, using a different match procedure, as described below. For this reason, and as a check on the coding of articles as direct replications, the 237 articles identified by Liu et al. (2025) was our initial set (see list at <https://osf.io/xmdv7>). This set was then augmented by a search in Web of Science of the 47 journals for additional Registered Reports published between April 1, 2023 and April 1, 2026 (see <https://osf.io/4vbtq/> for search and screening details). After screening, this led to an additional 111 articles. All articles included in this combined set ($N = 348$) set will be examined by the authorship team to ensure they meet the criteria of a Registered Report given previously⁵.

From this final set we will randomly select 100 articles for further analysis. Using these 100, we will locate companion articles for the 100 preregistered and 100 traditional articles. The total sample of articles will be 300 across the three types, which is informed by our power analysis (described in the subsequent section).

Companion articles will be selected based on the following criteria, modified from the criteria used by O'Mahony (2023).

Journal - the companion articles must be selected from the same journal as the Registered Report.

Timeframe - The companion articles should be selected from the same issue as the Registered Report. If no companion article is available in the same issue, we will examine the prior issue. If none is available in the prior issue, we will examine the subsequent issue. We will continue this process of alternately looking through the next prior and subsequent issues until a companion is identified that was published in the same year as the Registered Report.

Topic - The companion articles should generally be the same substantive topic of the Registered Report, broadly conceived. *Topic* will be coded as 1 - very similar (e.g., both focused on risk factors related to depression), 2 - mostly similar (e.g., both focused on risk factors for psychopathology), or 3 - somewhat similar (e.g., both are related to psychopathology). Ideally, all included articles would have a rating of 1 or 2, but 3s will be acceptable if needed to ensure that we select articles published in the same year as the Registered Report.

⁴ For accuracy, it is important to note that Sassenhagen and Bornkessel-Schlesewsky (2015) is the first Registered Report published in *Cortex*, a journal that pioneered the format, but that a special issue of the journal *Social Psychology* published in 2014 (Nosek & Lakens, 2014), one year earlier, consisted of 15 Registered Reports that were all replications (and thus were excluded from Liu et al. (2025)).

⁵ Because of the compressed timeline for completion of this project, this screening process was begun prior to registration.

Design - The companion articles should generally be the same design, for example experimental, intervention, observational, or longitudinal. *Design* will be coded as 1 - very similar, 2 - somewhat similar, or 3 - not similar. Ideally, all included articles would have a rating of 1 or 2, but 3s will be acceptable if needed to ensure that we select articles published in the same year as the Registered Report.

Exploratory Article Selection Process

We will include three additional sets of articles for exploratory analyses and assessments of generalizability:

First, an additional group of 100 traditional articles published in 2010 will be selected from the same journals as the Registered Reports. Starting with the first issue published in 2010, we will identify an empirical article that matches the previously selected companion traditional articles based on *Topic* and *Design*, using the same criteria described above. The selected articles must have a score of at least 2 for each of *Topic* and *Design* to be included.

Second, although Registered Reports that include a mix of registered and non-registered studies are excluded from the primary analyses, all articles coded as these “partial” Registered Reports will be coded separately at the study level, using the same coding process as the primary analyses, to determine any possible within-article variations in exploratory research based on study type. This analysis will only be based on Registered Reports, and thus inferences will be limited to that context. Because these articles will be identified through the coding process, the sample size is unknown.

Third, to assess the implications of using journals that publish Registered Reports as our starting point for article selection, we will select an additional sample ($n = 25$) of traditional articles from journals that do not publish Registered Reports. These journals will be selected from the Clarivate Journal Citation Report, prioritizing journals with the closest journal impact factor within the same category to the target journal as a way to match scope. Once the journals are selected, a random sample of 25 of the 100 traditional articles in the primary set will be selected. These 25 articles will be the basis for finding a match in non-Registered Report journals, based on the criteria for matches described above.

Analysis of Journal Guidelines

Journals differ greatly in both policies and practices related to open science, including standards for clearly and accurately labeling exploratory and confirmatory research. Assessing the direct impact of journal policies on article contents is difficult given that journal policies change over time, old policies are rarely archived, and it may not be possible to identify which articles were reviewed under specific policies. One way this will be addressed is by accounting for clustering by journal in the analysis, as described subsequently. Additionally, we will partially address the influence of journal policies by conducting an analysis of journal policies as indicated on the journal homepages in June 2026 (these will be archived at <https://osf.io/4vbtq/>). The policies will be coded for whether or not they include any guidance on distinguishing between exploratory and confirmatory analysis (required, suggested, or nothing), and whether this guidance is specified for Registered Reports, preregistered articles, and/or traditional articles.

Article Coding

The full detailed coding protocol is available at <https://osf.io/4vbtq/>. Here, we provide a summary of the main features of the protocol. Each article serves as the unit of analysis for coding, that is, the unit to which the coding categories will be applied.

Importantly, in the current project we remain agnostic about the quality of the registrations themselves. There have been several studies indicating that registrations lack specificity and that researchers often engage in undisclosed deviations from their registrations (Claesen et al., 2021; van den Akker et al., 2023). These are important issues but not our concern here, as we are focused on the implications of the practice of registration regardless of how that practice is implemented, and to what degree (if any) it restricts reporting of exploratory research.

Each article will be coded for the following:

Statements of Exploratory Research

One of the limits of the O'Mahony (2023) study is that it did not include a clear definition of what constitutes "exploratory." In the current project, the presence of exploratory research will be coded in accordance with our definition, "any analysis that is indicated to be exploratory in nature or indicated to not be a planned test of a specific hypothesis." We will code for any explicit or implied mention of exploratory tests, exploratory analysis, unplanned analysis, or speculative analyses. "Implied" is a subjective criterion but is included here for cases where raters can determine that the analyses were exploratory absent an explicit statement. For example, if researchers state a research question, but indicate that they made no hypotheses about that question, then that would be considered exploratory even if they did not explicitly label it as such. Additionally, any tests that are labeled as robustness tests, sensitivity tests, or tests of alternative explanations, and are explicitly or implied to be exploratory or unplanned, will be coded as exploratory research. In this way, we are not limited only to direct reporting (i.e., labeling) of research as exploratory, but rather any work that can be reasonably discerned as exploratory based on the information provided in the article.

Importantly, not all uses of the word "explore" or "exploratory" necessarily indicate exploratory research. For example, following a hypothesized statistically significant interaction effect, authors may state that they "explored" or "probed" the interaction via simple slopes analysis. That kind of follow-up test is directly related to the hypothesis about the interaction and is conducted to gain greater clarity on the specific nature of the interaction. It also reflects the broader problem in psychology with the lack of specificity when proposing hypotheses about interaction effects (Baranger et al., 2023). An additional case that does not necessarily meet the criteria for exploratory research is the use of "exploratory factor analysis" as an analytic technique. This type of analysis is sometimes used in an exploratory way that is consistent with our definition, and sometimes used to test confirmatory hypotheses (despite its name). In both of these cases, the type of analysis could reflect exploratory research by our definition, but it will depend on the specific use in that study. Thus, the context will be taken into account when coding rather than simply examining the articles for keywords.

All sections of each article will be inspected and coded for the presence of exploratory research: 1) Title; 2) Abstract; 3) the end of the Introduction section, where the researchers discuss the aims of the study; 4) Method section(s); 5) Results sections(s); and 6) the Discussion section(s).

Data Analysis Plan sections are not standard in the literature, and could constitute a separate section, a subsection of the Method or Results sections, or could be omitted altogether. They will be coded in the section in which they appear. If it constitutes a separate section, then it will be considered as part of the Method section. For articles that contain multiple studies, the multiple sections that serve the same function (e.g., Results for Study 1, Results for Study 2) will be treated as one, as all studies will be registered or not registered. For each section, raters will code for the presence or absence of mentions of exploratory research. Although O'Mahony (2023) also included an "uncertain" category, our criteria is "explicit or implied" references to exploratory analyses, therefore articles that would potentially be coded as uncertain would not meet our coding criteria and thus should be coded as absent.

The coding will be used to create two indices. *Presence* of exploratory research will be a binary (0 = no, 1 = yes) category collapsed across the six coded sections of each article that will correspond to any inclusion of exploratory research. *Prominence* of exploratory research will be a count variable (0-6) that sums across the six coded sections, and therefore corresponds to how much exploratory research is represented across the article.

Section Headers Indicating Exploratory Research

Raters will code presence/absence for the inclusion of section headers that explicitly indicate exploratory analyses. The full articles will be examined, but these are most likely to appear in the Results section when present.

Coding Difficulty

Raters will code for how difficult it was to determine their rating, which also can serve as a useful indicator of their confidence in their ratings. We will use the system from O'Mahony (2023): 1 - Very easy, 2 = Somewhat easy, 3 = Somewhat difficult, and 4 = Very difficult. Raters will provide a global rating for difficulty, rather than for each section.

Rater Training and Reliability

The raters for this project will consist of the four authors, a professor of psychology and three doctoral students in social psychology. We conducted a pilot coding process of six articles based on the preliminary coding categories. This pilot process indicated that the coding system was sufficiently clear and applicable, and led to some minor refinements in the definition and scope of the categories.

Formal training and establishing reliability will be completed after registration. The training phase will use articles that are not part of the focal set but share similar characteristics (i.e., articles of the three types that were published in psychology journals across the same time range). We will draw from the following journals that were not included in the focal set: *Assessment*, *Clinical Psychological Science*, *Journal of Personality*, *PloS ONE*, and *Social Psychological Bulletin*. These are journals drawn from diverse subfields of psychology and all currently publish Registered Reports, and thus are comparable to our focal set of journals.

The training phase will consist of four steps:

Step 1 will be for the rating team to become familiar with the coding system and how to apply it to the articles. We will independently code two articles during a project meeting and then have a discussion about our coding system.

Step 2 will involve the raters independently coding a set of five articles. This coding will be subject to reliability analysis using average pairwise percent agreement and Fleiss's Kappa to quantify the degree of agreement. Reliability will be examined for each of the eight components to be coded (six sections of the articles, section headers, and coding difficulty). The raters will then have a meeting in which they discuss the disagreements.

Step 3 will be a repeat of Step 2 but with 10 new articles. Once again, average pairwise percent agreement and Fleiss's Kappa will be used to quantify the degree of agreement. If percent agreement exceeds .80 and Kappa exceeds .60, then the training phase will be complete. If not, then we will continue the training phase with a new set of 10 articles until these thresholds are reached.

The focal set of 300 articles will be coded by the four raters, arranged in two teams of two raters each. Each team will code 125 unique articles, with the remaining 50 articles being coded by all four raters. The articles coded by all four raters will be interspersed across the coding period to assess rater drift. Percent agreement will be assessed for each batch, and the raters for that batch will meet to resolve discrepancies via discussion. All data and code will be made openly available at <https://osf.io/4vbtq/>.

Planned Analysis

Hypothesis Tests

Each hypothesis will be tested using three indices: presence, prominence, and section headings. Analytic code using simulated data is available at <https://osf.io/4vbtq/>. Alpha is set at .05, but we will use the false discovery rate adjustment (Benjamini & Hochberg, 1995) using the `p.adjust()` function in R to account for multiple tests. All analyses will be conducted in R version 4.5.2 (R Core Team, 2022).

Regarding probable effect sizes, the arguments made in the literature that registration will restrict exploratory research has often used extreme language such as that it will “put researchers in chains” (Scott, 2013), “stifle discovery” (Goldin-Meadow, 2016) and serves as a “stranglehold” on research (McDermott, 2022). The severity of this language clearly implies a large effect, that we would see a dramatically lower rate of exploratory work in registered compared to non-registered studies. The only available evidence, however, indicates a moderate difference in the *other* direction, with a greater prevalence of exploratory research in Registered Reports compared with standard reports (O'Mahony, 2023). The effect size phi, appropriate for 2x2 chi-square tests, was not reported in O'Mahony (2023), but we can use the reported test statistic, $\chi^2(1, N = 510) = 16.19, p < 0.001$, to calculate it as $\phi = .18$.

It is difficult to say what effect size is the smallest that would be meaningful. Our goal in this project, however, is to address the extreme claims about the damaging impact of registration, and thus are primarily interested in large effects. Given that we are conducting a more tightly controlled analysis than O'Mahony (2023), in terms of assessment of reported exploratory research, we opted

to use $\phi = .20$ as the target effect size for 2x2 chi squares and the corresponding Cohen's $d = .40$ for t -tests. Power analyses were conducted using the `pwr.chisq.test()` and `pwr.t.test()` functions from the *pwr* package in R (Champely et al., 2020). Power analysis based on $\phi = .20$, power = .80, and alpha = .05 indicated a total sample size of 196 articles. Power analysis based on $d = .40$, power = .80, and alpha = .05 (two-sided) indicated a sample size of 99 for each group.

All analyses will account for journal-level clustering. However, the power analysis does not consider clustering because it is unknown a) how many journals will be included in the final analysis, b) how many articles will come from each included journal, and c) how much journal-level variation to expect. As sensitivity tests, all analyses will also be conducted without accounting for clustering by journal.

Hypothesis 1: There will be a higher prevalence of reported exploratory research in preregistered articles compared to traditional articles.

Test 1, Presence: This test will be conducted using a 2x2 chi-square test of independence, comparing article type (preregistered, $n = 100$ or traditional, $n = 100$) with the binary outcome of presence/absence of exploratory research. The `chisqtestClust()` function from the *htestClust* package (Gregg et al., 2022) will be used to account for journal-level clustering.

Test 2, Prominence: This test will be conducted using an independent samples t -test. Article type (preregistered, $n = 100$ or traditional, $n = 100$) will be entered as a binary predictor of the mean count (0-6) of exploratory research across the six sections. The `ttestClust()` function from the *htestClust* package will be used to account for journal-level clustering. Exploratory follow-up tests will examine variations in the specific sections (e.g., Introduction vs. Results sections), but no inferential tests will be used and no p -values will be reported for these analyses.

Test 3, Section headings: This test will be conducted using a 2x2 chi-square test of independence, comparing articles type (preregistered, $n = 100$ or traditional, $n = 100$) with the binary outcome of presence/absence of section headings. The `chisqtestClust()` function from the *htestClust* package will be used to account for journal-level clustering.

Hypothesis 2: There will be a higher prevalence of reported exploratory research in Registered Reports compared to traditional articles.

Test 1, Presence: This test will be conducted using a 2x2 chi-square test of independence, comparing article type (Registered Report, $n = 100$ or traditional, $n = 100$) with the binary outcome of presence/absence of exploratory research. The `chisqtestClust()` function from the *htestClust* package (Gregg et al., 2022) will be used to account for journal-level clustering.

Test 2, Prominence: This test will be conducted using an independent samples t -test. Article type (Registered Report, $n = 100$ or traditional, $n = 100$) will be entered as a binary predictor of the mean count (0-6) of exploratory research across the six sections. The `ttestClust()` function from the *htestClust* package will be used to account for journal-level clustering. Exploratory follow-up tests will examine variations in the specific sections (e.g., Introduction vs. Results sections), but no inferential tests will be used and no p -values will be reported for these analyses.

Test 3, Section headings: This test will be conducted using a 2x2 chi-square test of independence, comparing articles type (Registered Report, $n = 100$ or traditional, $n = 100$) with the binary outcome of presence/absence of section headings. The `chisqtestClust()` function from the *htestClust* package will be used to account for journal-level clustering.

Hypothesis 3: There will be a higher prevalence of exploratory research reported in Registered Reports compared with preregistered articles.

Test 1, Presence: This test will be conducted using a 2x2 chi-square test of independence, comparing article type (Registered Report, $n = 100$ or preregistered, $n = 100$) with the binary outcome of presence/absence of exploratory research. The `chisqtestClust()` function from the *htestClust* package (Gregg et al., 2022) will be used to account for journal-level clustering.

Test 2, Prominence: This test will be conducted using an independent samples *t*-test. Article type (Registered Report, $n = 100$ or preregistered, $n = 100$) will be entered as a binary predictor of the mean count (0-6) of exploratory research across the six sections. The `ttestClust()` function from the *htestClust* package will be used to account for journal-level clustering. Exploratory follow-up tests will examine variations in the specific sections (e.g., Introduction vs. Results sections), but no inferential tests will be used and no *p*-values will be reported for these analyses.

Test 3, Section headings: This test will be conducted using a 2x2 chi-square test of independence, comparing articles type (Registered Report, $n = 100$ or preregistered, $n = 100$) with the binary outcome of presence/absence of section headings. The `chisqtestClust()` function from the *htestClust* package will be used to account for journal-level clustering.

Exploratory Analyses

We will conduct extensive exploratory analyses of the data, focusing on generating plots and reporting effect sizes. No inferential statistics will be used and no *p*-values will be reported. Planned exploratory analyses include: 1) Within-article analyses of the frequency of exploratory research reported within mixed registered and non-registered articles, 2) comparison of traditional articles published in 2010 to traditional articles included in the primary set, after the journals started publishing Registered Reports, 3) changes over time in reporting of exploratory research across the three article types, and 4) relations between journal policies and reporting of exploratory research.

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